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Study on cerebroprotective actions of *Clerodendron glandulosum* leaves extract against long term bilateral common carotid artery occlusion in rats



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ABSTRACT

Stroke is a major cause of death and disability worldwide. The resulting burden on the society continues to grow, with increase in the incidence of stroke. Oxidative stress has been involved in the pathogenesis of several neurological diseases including acute stroke. Focal and global cerebral ischemia represents diseases that are common in the human population. In recent years much attention is being paid towards the exploration of herbal preparation, antioxidant agents and combination therapies including COX-2 inhibitors in experimental model of stroke. Possible effect of a hydroalcoholic leaf extract of *Clerodendron glandulosum* (C. *glandulosum*) on oxidant-antioxidant status in ischemia-hypoperfusion injury in the rat forebrain has been investigated. Healthy adult male Wistar albino rats were divided into five groups ($n=8$). Group I was served as Sham control (normal saline 1 ml/kg, orally), group II was served hypoperfusion control (normal saline 1 ml/kg, orally), group III, group IV were served as hydroalcoholic extract treated (200 and 400 mg/kg, orally) and group V was treated with Quercetin (10 mg/kg, orally) for 14 days to assess preventive and curative effects of *C. glandulosum*. Flavonoid and phenolic compounds exhibit a broad spectrum of biological activity, including antioxidant. *C. glandulosum* extract (200 and 400 mg/kg, p.o) was administered orally, once daily for a period of 2 weeks after the occlusion of BCCA. After 14th days rats were subjected to behavioral studies. After behavioral studies animals were sacrificed and brain was removed and homogenized. Estimation of Lipid peroxidation (LPO) Myeloperoxidase (MPO), estimation of protein levels and the activities of Superoxide dismutase (SOD), Catalase (CAT), were performed. Infarct size and histopathological changes were observed in treated groups.

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1. Introduction

Cerebral ischemia is a condition in which there is insufficient blood flow to the brain to meet metabolic demand. This leads to poor oxygen supply or cerebral hypoxia and thus to the death of brain tissue or stroke [1]. Among stroke patients, 80% suffer from cerebral infarction and 20% from cerebral hemorrhage [2]. Risk factors for stroke include advanced age, hypertension, transient ischemic attack, diabetes; high cholesterol, cigarette smoking and atrial fibrillation. Many antioxidants are reported to reduced reactive oxygen species-mediated reaction and protect neurons from ischemia-reperfusion induced neural loss in animal models of cerebral ischemia. Methanolic extract of *C. glandulosum* leaves

extract was assayed for its qualitative and quantitative phytochemical constituents as well as for its exhibits free radical scavenging potential using different *in vitro* assays for hydrogen peroxide and hydroxyl superoxide, DPPH, nitric oxide, peroxytrite, singlet oxygen and hypochlorous acid radicals. It also exhibits hepatoprotective, anti-obesity, anti-atherosclerosis, prevention of metabolic syndromes, non-alcoholic steatohepatitis activity [3]. Traditionally, people across Manipur consume decoction of *C. glandulosum* leaves for treating diabetes, obesity and hypertension. Phytochemical analysis of extract of *C. glandulosum* revealed presence of phenols, steroids, alkaloids, anthraquinones, flavonoids and polyphenols. The hydroalcoholic extract of the leaves showed antioxidant activity [3]. There is considerable evidence which supports the role of reactive oxygen species (ROS) in the pathogenesis of ischemic reperfusion induced oxidative stress in brain were reported [4]. Novel therapeutic neuroprotective strategies support the applications of ROS scavengers and

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induction of endogenous antioxidants drugs, such as natural antioxidants, e.g. plant derived polyphenolic compounds, flavonoids, in mono therapy, or as part of an antioxidant formulation for the treatment of neurodegenerative diseases [5]. Hence on the basis of the literature review, and active constituents present in the leaves of *C. glandulosum*, this study was undertaken to study the cerebroprotective effect of hydroalcoholic extract of *C. glandulosum* against bilateral common carotid artery occlusion induced cerebral ischemia in rats.

2. Materials and methods

2.1. Animals

After obtaining permission from Institutional Animal Ethics Committee (IAEC) with study protocol number (IAEC/ABMRCP/2012-13/01), male albino Wistar rats (150–250 g) were obtained from Bioneed, Bangalore, Karnataka. All the experiments were performed in accordance with the guidelines of Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA). Rats were caged in fully ventilated room. Animals were maintained in 12:12 h light and dark cycle and were housed at temperature of $25 \pm 2^\circ\text{C}$. They had free access to a standard chow diet (Chow diet contain ingredients such as ground corn, ground oats, alfalfa meal, soybean meal and ground wheat, as well as vitamins, minerals and fat are added to ensure nutritional adequacy) and water *ad libitum*.

2.2. Chemical

Nicotinamide adenine dinucleotide phosphate reduced (NADPH), 1-chloro-2,4-dinitrobenzene (CDNB), thiobarbituric acid (TBA), O-dianisidinedihydrochloride, 5-5-dithiobis-2-nitrobenzoic acid (DTNB), Folin-Ciocalteu phenol reagent, ethylene diamine tetra acetic acid (EDTA), nitrobluetetrazolium (NBT), 2,3,5-triphenyltetrazolium chloride (TTC), Quercetin, were procured from Sigma Aldrich (St. Louis, MO, USA) and SRL, Bombay, India.

The plant was authenticated by Dr.T.H. Brojendra Singh, Institute of Biotech Hub Centre, Takyalapat, Imphal, Manipur. The voucher specimen no-1BT H-OC-01-2013 of the plant was deposited in the department. All other chemicals and reagents used were of analytical grade and procured from approved vendors.

2.3. Preparation of extract

The plant, *C. glandulosum* leaves were procured from Manipur India. The leaves of *C. glandulosum* were shade dried. Dried leaves were then powdered and passed through sieve no: 44. Fine powder was collected; 100 g of fine powder was packed, and the powdered material was defatted with petroleum ether, chloroform solvents. The defatted powder was extracted with hydroalcoholic solvents at $65\text{--}70^\circ\text{C}$ for 24 h in soxhlet column. After extraction the solvent was removed by using vacuum evaporator. Dried extract was stored in bottle wrapped with aluminium foil at $2\text{--}8^\circ\text{C}$. The chemical constituents of the hydroalcoholic extract of *C. glandulosum* were identified by qualitative analysis [6].

2.3.1. Determination of total flavonoid compound

The total flavonoid compounds in extract were determined by an aliquot (1.5 ml) of hydroalcoholic extract of *C. glandulosum* was added to 6 ml of deionized water and then 0.45 ml of 5% (w/v) NaNO_2 and incubated for 6 min. AlCl_3 10% (w/v) of 0.45 ml and 6 ml of 4% (w/v) NaOH was added and the total volume was made upto 15 ml with distilled water. The absorbance was measured at 510 nm by using visible spectrophotometer. The results were

expressed as milligram quercetin equivalent/g. The experiments were performed by triplicates [6].

2.3.2. Determination of total phenol compound

Total phenolic compounds present in the hydroalcoholic extract of *C. glandulosum* were determined with the Folin-Ciocalteu reagent method. 1 ml of the sample solution was mixed with 1 ml Folin-Ciocalteu reagent. After 3 min, 35% of 3 ml Na_2CO_3 was added to the mixture followed by the addition of 7 ml of distilled water. The reaction mixture was kept in dark for 90 min and the absorbance was measured at 725 nm. The concentration of the total phenolic compound was determined as milligram gallic acid equivalent/g extract [6].

2.4. Acute toxicity

Acute toxicity studies were carried out by using rats as per Organization for Economic Co-operation and Development (OECD) 425 guidelines and the protocol was approved by Institutional Animal Ethics Committee (IAEC) (Protocol No: IAEC/ABMRCP/2013-14/01). Mice were divided into three groups and three animals in each group. A limit test at dose of 2000 mg/kg body weight was carried out of hydroalcoholic extract. The animals were observed for clinical signs and mortality for 15 days. Body weight was recorded every week. The tested sample was found to be safe and did not produce any mortality after 15 days. [7]

2.5. Experimental design

The study was planned to assess ability of *C. glandulosum* to cure ischemic injury due to cerebral ischemia. Wistar albino rats were randomly divided into five groups containing eight animals each. In Group I (Sham control) cerebral ischemia was not induced. Group II (hypoperfusion control) cerebral ischemia was induced and treated with normal saline (1 ml/kg, p.o) for 14 days. Group III & Group IV were induced cerebral ischemia and treated with *C. glandulosum* (200 & 400 mg/kg, p.o) for 14 days. Group V was induced cerebral ischemia and treated with Quercetin (10 mg/kg, p.o) for 14 days as reference group.

2.5.1. Induction of cerebral ischemia

Surgical technique for induction of cerebral ischemia was adapted from the method of Iwasaki et al. (1989) and rats were anesthetized by giving thiopentone sodium (40 mg/kg, i.p.). Both common carotid arteries were exposed over a midline incision, and a dissection was made between the sternoicloidomastoid and the sternohyoid muscles parallel to the trachea. Each carotid artery was freed from its adventitial sheath and vagus nerve, which was carefully separated and maintained. A cotton thread was passed below each carotid artery. The induction of the ischemia was performed by BCCA occlusion with cotton thread. Temperature was maintained around $37 \pm 0.5^\circ\text{C}$. The carotid arteries were doubly ligated with 3-0 silk thread and cut in between. The skin was then sutured, animals were kept in the individual cages and applied betadine ointment by sterilized cotton for sutured animals. The dose of *C. glandulosum* was continued up to 14th post surgical day. On the 14th day, 60 min after last dose of *C. glandulosum*, the animals were subjected to behavioral study [8].

2.6. Effect of *C. glandulosum* on behavioural parameters

2.6.1. Open field test

The locomotor activity was evaluated in an open field paradigm. The apparatus consisted of a wooden box ($60 \times 60 \times 30$ cm). The floor of the box was divided into 16 squares (15×15 cm). The apparatus was illuminated with a 40-W lamp suspended 100 cm

above. Each animal was placed at one corner of the apparatus and for the next 5 min it was observed for the ambulation (number of squares crossed), total period of immobility (in seconds), number of rearing's, grooming, faecal pellets [9].

2.6.2. Elevated plus maze test

The elevated plus maze test (EPM) consisted of two open arms (35 × 5 cm) crossed with two closed arms (35 × 5 × 20 cm). The arms were connected together with a central square of 5 × 5 cm. The apparatus was elevated to the height of 50 cm in a dimly illuminated room. Animals were placed individually at the end of either of the open arms facing away from the central platform. The time taken by each animal to move from open arm to either [10].

2.6.3. Despire swim test

The rats were placed individually in a cylinder (45 × 20 cm) containing 38 cm water (25 ± 2 °C), so that the rats could not touch the bottom of the cylinder with its hind limbs or tail or climb over the edge of the chamber. Two swim sessions were conducted, an initial 15 min pre-test followed by a 5 min test, 24 h later. *C. glandulosum* extract was administered after pre-test. The period of immobility (duration of floating in water without struggling and making only those movements necessary to keep its head above the water level), during 5 min test period was noted [11].

2.6.4. Morris water maze test

The test for learning and memory on day 15 after surgery, spatial learning and memory was tested in water maze. The maze consists of a black circular pool (diameter 2.14 m, height 80 cm) filled to a depth of 44 cm with water (25 °C). On 14th day the rats received habituation (exposure in water maze for 1 min) in which there is no platform present. Then, on day 15th, a circular platform (9 cm in diameter) was kept hidden 2 cm below water level in the centre of one of the quadrants. The platform remains in the same position during training days. At the beginning of each session, a random sequence of four starting poles along the perimeter of the pool was generated. All animals followed this sequence for that session. Each rat was placed in the water facing the wall at the start location and was allowed 90 s to find the hidden platform. The animal was allowed a 20 s rest on the platform. The latency to reach the platform was recorded. If the rat was unable to locate the hidden platform, it was lifted out and placed on the platform for 20 s. The procedure was repeated for all the four start locations. Two sessions of four trials each was conducted on the first day of testing separated by 4 h and one session of four trials was conducted on the next day. After that, the platform was removed and a probe trial (without platform) was conducted 4 h later. Each rat was placed in the pool at the same randomly selected starting pole and swimming path was observed and time spent in the quadrant of pool which initially contained platform was measured. On completion of the probe trial, a black platform that extends 1 cm above the surface of water was placed in a quadrant other than that chosen for the submerged platform. Each rat was then given four trials of 90 s to locate it. The latency to reach the platform was recorded [12].

2.7. Preparation of brain homogenate

After behavioral studies, hypoperfusion animals were decapitated. The brain was removed and washed in ice chilled 0.9% saline and subsequently blotted on filter paper. The brain was weighed and homogenized in cold phosphate buffer (0.1 M, pH 7.4) to make a 10% w/v homogenate using a homogenizer. The homogenate was centrifuged at 10,000 rpm for 20 min at 4 °C. Separated the supernatant and was kept in deep refrigerator at –80 °C for biochemical study [13].

2.8. Biochemical estimation

2.8.1. Superoxide dismutase

SOD activity was determined by adding the 25 µl of the supernatant obtained from the centrifuged brain homogenate 1.2 ml of 0.052 M sodium pyrophosphate buffer (pH 8.3) followed by addition of 0.1 ml of 186 µM phenazoniummetho sulphate, 0.3 ml of 300 µM of nitrobluetetrazolium, 0.2 ml of 780 µM NADH. Reaction mixture was incubated for 90 s at 30 °C, and the reaction was stopped by the addition of 0.1 ml of glacial acetic acid. Reaction mixture was stirred vigorously and shaken with 4.0 ml of *n*-butanol and centrifuged at 4000 rpm for 10 min. Superoxide dismutase is an enzyme that catalyzes the dismutation of superoxide into oxygen and hydrogen peroxide. A control was prepared by using 0.1 ml of distilled water of 0.1 ml of homogenate [15].

2.8.2. Catalase

Catalase activity was measured by adding supernatant 0.1 ml of brain tissue was added to cuvette containing 1.9 ml of 50 mM phosphate buffer. To this mixture, 1.0 ml of freshly prepared 30 mM H₂O₂ was added and changes in absorbance for 3 min at 240 nm at an interval of 30 s was measured. A control was prepared using 0.1 ml of distilled water devoid of 0.1 ml of homogenate [16].

2.8.3. Malondialdehyde

Lipid peroxidation end product Malondialdehyde (MDA) was measured by adding the 0.2 ml of brain homogenate was treated with 0.2 ml of sodium dodecyl sulphate (8.1%), 20% of 1.5 ml of acetic acid (pH 3.5), 1.5 ml of thiobarbituric acid (0.8%). The mixture was made up to 5 ml with distilled water and then heated at 95 °C in oil bath for 60 min. The mixture was cooled and 5 ml of *n*-butanol and pyridine mixture (15:1 v/v) was added. The mixture was shaken vigorously. After centrifugation of the mixture at 4000 rpm for 10 min, the organic layer was taken and the absorbance was measured at 532 nm [14].

2.8.4. Myeloperoxidase

Myeloperoxidase (MPO) was measured by the method of Mullanet *al.* the Supernatant 0.1 ml of brain tissue was added 0.5% hexadecyltrimethyl ammonium bromide. Phosphate buffer (960 µl) containing O-dianisidinedihydrochloride (0.167 mg/ml) and hydrogen peroxide (0.0005%) solution and shaken vigorously. The change in the absorbance was measured at 460 nm for 3 min at an interval of 60 s. One unit of activity was defined as the amount of MPO that causes a change in absorbance measured at 460 nm for 3 min MPO activity data are presented as units/ml [17].

2.8.5. Protein

Estimation of protein was performed by protein concentration in all PMS 10% (w/v) sample determined by Lowry et al. [18].

2.9. Infarct size

After behavioral studies, animals were decapitated and the brain was removed and frozen at –4 °C for 5 min. Coronal slices were made at 1 – 2 mm and sections were immersed in 1% of 2, 3, 5-triphenyltetrazolium chloride (TTC) at 37 °C for 20 min. TTC is converted to red formazone pigment by nicotinamide adenine dinucleotide (NAD) and dehydrogenase present in living cells. Hence, viable cells were stained deep red. The infarcted cells have lost the enzymes and thus remained unstained. Whole brain slices were weighed. Infarcted unstained part was dissected out, weighted and expressed as % of total weight of brain [19].

Table 1Effect of *C. glandulosum* on open field test in hypoperfusion on BCCA occlusion in rats.

Treatment groups	Dose (mg/kg, p.o.)	Ambulation's (number)	Immobility (s)	Grooming (number)	Rearing (number)	Faecal pellet (number)
Sham	Normal saline	55.80 ± 6.05	170.80 ± 33.50	6.40 ± 1.14	7.00 ± 1.22	2.60 ± 1.14
Hypoperfusion control	Normal saline	28.50 ± 3.31	131.75 ± 31.42	4.75 ± 1.70	3.50 ± 2.64	3.75 ± 0.50
<i>C. glandulosum</i>	200	34.25 ± 6.02 ^{ns}	167.00 ± 48.14 ^{**}	6.50 ± 2.38	4.25 ± 1.25	4.50 ± 0.57
<i>C. glandulosum</i>	400	47.75 ± 3.86 ^{**}	165.25 ± 16.33 ^{**}	7.50 ± 1.73	6.75 ± 1.70 ^{**}	4.75 ± 1.25 [*]
Quercetin	10	38.00 ± 4.96	202.00 ± 12.72	3.25 ± 0.95 ^{***}	2.50 ± 1.29 ^{**}	2.00 ± 0.81 [*]

All values are expressed as Mean ± SD. Data was analyzed by One-Way Analysis of Variance (ANOVA) followed by Dunnett's test. **P* < 0.05, ***P* < 0.01 and ****P* < 0.001 as compared to hypoperfusion control group.

2.9.1. Histopathological examinations

At the end of experimental procedures and behavioral testing, rats were sacrificed and the brain were taken out. The brains were transferred to 10% formalin. Sections (< 5 μm thick) of the frontal forebrain were prepared and stained by hematoxylin and eosin for microscopical examination [20].

2.10. Statistical analysis

The results were expressed as (Mean ± SD). MDA, SOD, CAT, MPO and behavioral parameters were determined by factorial one-way ANOVA. Individual group was compared against control using Dunnett's test. Values ranging *P* < 0.001 were considered statistically significant. Statistical analysis was performed using Prism software (Version 5.01).

3. Results

3.1. Phytochemical screening

Preliminary phytochemical screening revealed that hydro-alcoholic extract showed positive for the presence of flavonoids, steroids, phenols and polyphenols, carbohydrates, alkaloids, fats and oils, saponin glycosides. Yield of the dried powder hydro-alcoholic extract was 21.1% w/w and total phenols (gallic acid equivalents) and flavonoids (quercetin equivalents) contents from the hydroalcoholic extract of leaves were found 8.75 and 8.5 mg/g extract.

3.2. Effect of *C. glandulosum* on behavioral studies

Effect of *C. glandulosum* on elevated plus maze test, despire swim test, open field test, morris water maze test on rat's results are shown in Table 1–4.

3.3. Open field test

Animals with permanent BCCAO (hypoperfusion group) showed marked alterations in locomotor activities in open field paradigm. Permanent BCCAO was associated with reduced number of ambulation (28.50 ± 3.31), rearing (3.50 ± 2.64) and grooming (4.75 ± 1.70) along with a decrease in the period of immobility (131.75 ± 31.42). *C. glandulosum* (400 mg/kg) treated group prevents the alteration of ambulation (47.75 ± 3.86), rearing (6.75 ± 1.70) and grooming (7.50 ± 1.73) along with an increase in the period of immobility (165.25 ± 16.33).

3.3.1. Elevated plus-maze test

A lower proportion of open arm entries was observed in the groups with decreased in enclosed arm entries in animals with hypoperfusion group (189.50 ± 75.97) as compared with the treatment of 200 mg/kg (228.75 ± 25.60) treatment also increased the proportion of time spent in the open arms while decreased the proportion of time spent in the enclosed arms.

The effect of *C. glandulosum* on the despire swim test performance and the more immobility (85.00 ± 10.39) in sec is observed the hypoperfusion group and shown less immobility (45.00 ± 4.54) at 400 mg/kg treated group.

Table 2Effect of *C. glandulosum* on elevated plus maze test in hypoperfusion on BCCA occlusion in rats.

Treatment groups	Dose (mg/kg p.o.)	Time spent (s)		No. of entries		Ratio of time spent (OA/CA)	Ratio of no. of entries (OA/CA)
		OA	CA	OA	CA		
Sham	Normal saline	273.80 ± 19.80	74.80 ± 60.50	5.00 ± 0.80	3.00 ± 0.30	0.581 ± 0.33	0.700 ± 0.96
Hypoperfusion control	Normal saline	189.50 ± 75.97	27.50 ± 25.90	2.65 ± 0.80	1.40 ± 1.00	0.170 ± 0.71	0.941 ± 0.91
<i>C. glandulosum</i>	200	228.75 ± 25.60	36.25 ± 24.90	2.75 ± 0.50	1.25 ± 0.95	0.150 ± 0.102	0.458 ± 0.32
<i>C. glandulosum</i>	400	205.00 ± 64.90	22.25 ± 0.95 [*]	3.75 ± 0.50 ^{**}	2.25 ± 0.95 ^{**}	0.125 ± 0.09 ^{**}	0.583 ± 0.20 ^{**}
Quercetin	10	283.25 ± 24.06 ^{ns}	28.25 ± 12.90 [*]	2.65 ± 0.95 ^{**}	1.75 ± 0.50 ^{**}	0.093 ± 0.05 ^{**}	0.512 ± 0.23 ^{**}

All values are expressed as Mean ± SD. Data was analyzed by One-Way Analysis of Variance (ANOVA) followed by Dunnett's test. **P* < 0.05, and ***P* < 0.01 as compared to hypoperfusion control group; ns = non-significant, CA = closed arm; OA = open arm.

Table 3Effect of *C. glandulosum* on despair swim test in hypoperfusion on BCCA occlusion in rats.

Treatment groups	Dose (mg/kg)	Immobility (s)
Sham	Normal saline	48.40 ± 12.60
Hypoperfusion control	Normal saline	85.00 ± 10.39
<i>C. glandulosum</i>	200	58.75 ± 10.50
<i>C. glandulosum</i>	400	45.00 ± 4.54***
Quercetin	10	54.50 ± 5.50***

All values are expressed as Mean ± SD. Data was analyzed by One-Way Analysis of Variance (ANOVA) followed by Dunnett's test. *** $P < 0.001$ as compared to hypoperfusion control group.

3.3.2. Morris' water maze test

Table 4 shows all rats were able to locate the hidden platform during the sessions of escape trial. The hypoperfused animals took longer time (6.0 ± 2.41) to locate the new platform than the *C. glandulosum* (400 mg/kg) treated animals took less time (4.0 ± 0.81) in finding the new visible platform.

3.4. Effect of *C. glandulosum* on anti-oxidant enzymes

The biochemical results are showed in Table 5. The results showed that the hypoperfusion significantly ($P < 0.001$) decreased antioxidant activities (SOD and CAT) and increased the level of MDA and MPO levels in the injured brain tissue of rats as compared with the sham control group. However, the pretreatment of rats with *C. glandulosum* (200 and 400 mg/kg) was markedly increased SOD and CAT, activity. In contrast, MDA and MPO content in the injured brain tissue of rats decreased significantly ($P < 0.001$) in *C. glandulosum* extract treated group compared to hypoperfusion control group.

3.5. Effect of *C. glandulosum* on infarct size

Significant increase in cerebral infarct size was measured by volume and weight method in hypoperfusion animals. *C. glandulosum* extract markedly decreased cerebral infarct size in a dose dependent manner as compared hypoperfusion control group (Fig. 1).

3.6. Effect of *C. glandulosum* on histopathological studies

Histopathological studies of the brain of hypoperfusion groups showed some pyramidal cells and neuroglial cells showing degenerative changes (necrosis). Few of the blood vessels show fibrinoid necrosis with features of vasculitis. Extract treated groups showed lesser changes in blood vessels as well as decreased necrosis and gliosis of congested blood vessels. No odema was observed in hypoperfusion animals (Fig. 2). *C. glandulosum*.

4. Discussion

The excessive generation of reactive oxygen species (ROS) and reactive nitrogen species (RNS) leads to oxidative or nitrosativa damage, or both, to cellular proteins, lipids, and DNA, which is collectively referred to as oxidative stress. Compared with other organs, the brain is more susceptible to oxidative damage due to its high metabolic rate, high polyunsaturated lipid content, and lower levels of endogenous antioxidant activity and protecting mechanisms. Ischemia and hypoperfusion causes brain injury via multiple pathways. Post ischemic reperfusion produces cell injury by inhibition of active ion transmembrane transport, depletion of energy rich adenosine triphosphate (ATP) stores within the ischemic tissues, calcium overload, arachidonic acid pathway disturbances, free radical generation and increased leukocyte endothelial cell and neutrophil interactions [21].

One of the most frequently used models of chronic cerebral hypoperfusion is bilateral permanent ligation of the common carotid artery of rats. In such a model hypoperfusion affects multiple brain regions, including the cortex, subcortical area, hippocampus, and others. The role of sensorimotor cortex and hippocampus is undisputable in memory and learning.

The reduction in total activity of the animals in the open field paradigm with significant reduction in the number of ambulation, Immobility, grooming, and rearing BCCA occlusion animals suggest a propensity towards anxiety and restlessness. *C. glandulosum* has significantly prevented ischemic reperfusion-induced anxiety and restlessness. The hypoperfused animals had deficits of spatial learning and memory as indicated by elevated plus maze test which is in accordance with earlier reports of ischemia-induced disturbances of spatial learning and memory [22]. Hypoperfused animals showed tendency towards depression identified by despair swim test and immobility time in *C. glandulosum* treated group was significantly decreased. The animals consistently showed increased transfer latencies suggesting a defective registration of the learning task indicated by Morris water maze test. *C. glandulosum* improve of learning and memory deficits by decrease in transfer latency.

The principal finding of the present study is that the cerebral ischemic reperfusion injury was associated with oxidative stress, as evidenced by increase in brain MDA and MPO level and depletion of cerebral endogenous antioxidant status (SOD and CAT). Similar observations were made earlier by other studies [23,24]. Increased levels of MDA reflect the membrane damage induced by toxic lipid peroxidation products. During cerebral ischemia, the circulatory and resident cells secrete inflammatory cytokines in the damaged areas and cause inflammation. MPO is stored in azurophilic granules of leukocytes and found in neutrophils, monocytes and macrophages. *C. glandulosum* exhibit a reduction of MDA and MPO levels in hypoperfusion groups.

Table 4Effect *C. glandulosum* on Morris water maze test in hypoperfusion on BCCA occlusion in rats.

Treatment groups	Dose (mg/kg, p.o.)	Escape latencies		Probe trial	New platform
		1	2		
Sham	Normal saline	7.60 ± 1.14	6.40 ± 1.94	5.80 ± 1.54	7.2 ± 2.30
Hypo control	Normal saline	5.00 ± 1.82	4.00 ± 8.16	3.50 ± 1.73	6.0 ± 2.41
<i>C. glandulosum</i>	200	7.25 ± 1.29	5.25 ± 1.29	4.25 ± 0.95	5.5 ± 0.57
<i>C. glandulosum</i>	400	7.50 ± 1.29*	5.75 ± 0.95	5.00 ± 0.81	4.0 ± 0.81
Quercetin	10	8.75 ± 0.95	6.00 ± 0.81	4.00 ± 0.81	4.5 ± 1.29

All values are expressed as Mean ± SD. Data was analyzed by One-Way Analysis of Variance (ANOVA) followed by Dunnett's test. * $P < 0.05$, as compared to hypoperfusion control group.

Table 5Effect of *C. glandulosum* extract on anti-oxidant enzymes in hypoperfusion on BCCA occlusion in rats.

Treatment Groups	Dose (mg/kg, p.o.)	SOD (Units/mg protein)	CAT (μ moles of H_2O_2 metabolized/mg protein/min)	MDA (nmol/gm wet tissue)	MPO (Units/ml)
Sham	Normal saline	3.816 $\pm$ 0.271	0.773 $\pm$ 0.189	0.884 $\pm$ 0.130	3.852 $\pm$ 0.717
Hypo control	Normal saline	0.796 $\pm$ 0.108	0.218 $\pm$ 0.130	2.868 $\pm$ 0.099	13.662 $\pm$ 2.192
<i>C. glandulosum</i> 200	200	1.100 $\pm$ 0.590	0.296 $\pm$ 0.059**	2.065 $\pm$ 0.170***	7.920 $\pm$ 0.400***
<i>C. glandulosum</i> 400	400	1.813 $\pm$ 0.168	0.542 $\pm$ 0.104 ***	1.334 $\pm$ 0.128***	8.013 $\pm$ 0.668***
Quercetin	10	1.882 $\pm$ 0.222***	0.569 $\pm$ 0.185***	1.063 $\pm$ 0.099***	5.208 $\pm$ 1.365**

All values are expressed as Mean $\pm$ SD. Data was analyzed by One-Way Analysis of Variance (ANOVA) followed by Dunnett's test. ** $P < 0.01$ and *** $P < 0.001$ as compared to hypoperfusion control group.

Superoxide anion is capable of producing significant biological injury. It is generated on reperfusion of post ischemic tissue [25]. Decrease in the levels of SOD was noted in brain of the ischemic rats that indicates participation of superoxide radical which known to produce highly toxic hydroxyl radical through its reaction with H_2O_2 [Haber–Weiss reaction [26]. Catalase is a common enzyme found in all living organisms which exposed to oxygen, where it has functions to catalyze the decomposition of hydrogen peroxide to water and oxygen. Decrease in the level of CAT was noted in brain of the ischemic rats. That indicates participation of superoxide radical which is known to produce highly toxic hydroxyl radical through its reaction with H_2O_2 [27,28].

The administration of *C. glandulosum* prevented the oxidative stress and the biochemical changes associated with cerebral ischemic hypoperfusion injury. The mechanism of such protection of oral administration of *C. glandulosum* may be due to cerebral adaptation, through augmentation of cellular antioxidants such as SOD and CAT. In cerebral ischemic hypoperfusion injury, oxidative stress plays a central role in its etiopathogenesis. Protection

against oxidative stress through this mechanism may be one of the effective therapeutic approaches.

Infarct size was measured by TTC method, TTC is converted to red formazone pigment by nicotinamide adenine dinucleotide (NAD) and dehydrogenase present in living cells. Hence viable cells were stained deep red. The infarcted cells loose these enzymes and thus remain unstained [29]. In this study we observed that treatment of *C. glandulosum* after BCCA occlusion could markedly decrease cerebral damage. Histopathological observations revealed that bilateral common carotid occlusion lead to changes like gliosis, congested blood vessels and cellular edema. *C. glandulosum* treatment in hypoperfusion models alleviated these histopathological changes.

The extract of *C. glandulosum* are known to contain flavanoids have been report to possess antioxidant property [3]. The observed beneficial effect of *C. glandulosum* hypoperfusion injury induced changes in biochemical parameters may they be attributed to its flavanoids content.

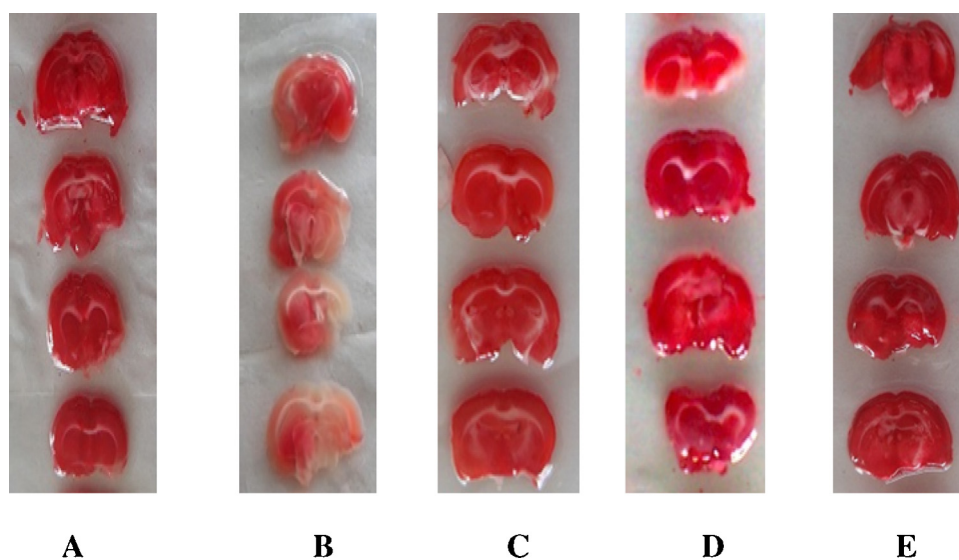


Fig. 1. Effect of hydroalcoholic extract of *C. glandulosum* on panel of coronal slices of the rat brain subjected to hypoperfusion in BCCA occlusion. The slices were stained with TTC. (A) Sham control shows deep red staining of normal brain tissue, (B) Hypoperfusion in BCCAO ischemic control shows the area of infarct (white color), (C) *C. glandulosum* (200 mg/kg, p.o.) shows red staining of brain tissue with more area of infarct, (D) *C. glandulosum* (400 mg/kg, p.o.), shows deep red staining of brain tissue with less area of infarct, (E) Quercetin (10 mg/kg, p.o.), shows deep red staining of normal brain tissue with no area of infarct (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

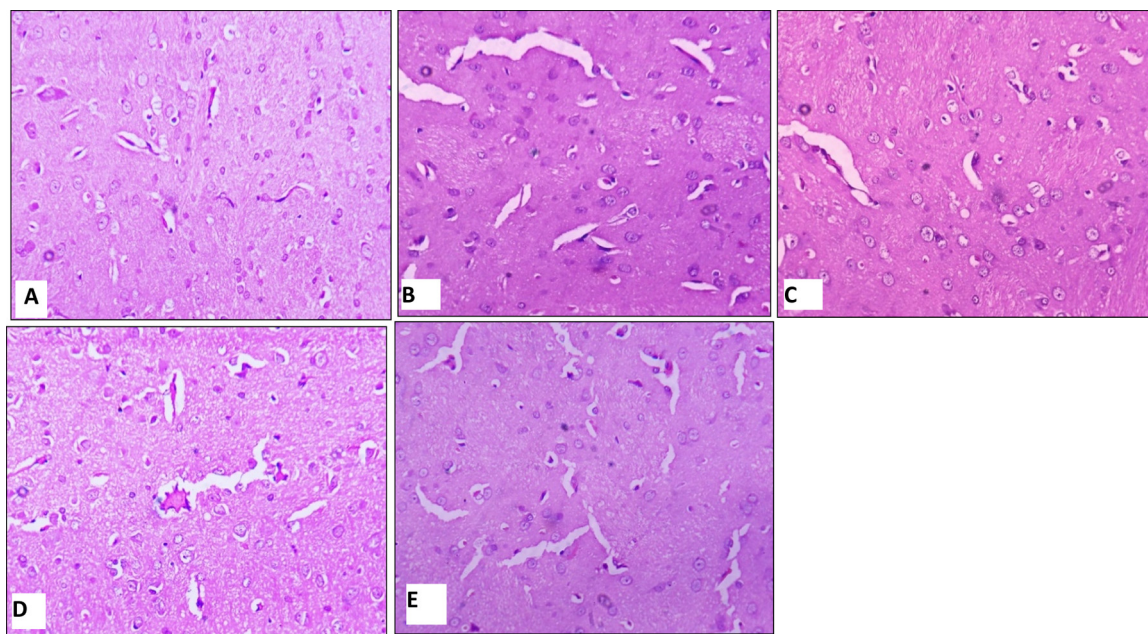


Fig. 2. Histopathological photographs of coronal sections of brain after hypoperfusion on BCCAO occlusion shown some pyramidal cells and neuroglial cells shown degenerative changes (necrosis) (A) Sham control shown normal brain tissue (B) Hypoperfusion in BCCAO ischemic control shown [Congested blood vessels (++++); Gliosis (++)]; Oedema (++++)] (C) *C. glandulosum* shown [Congested blood vessels (+++); Gliosis (+++); Oedema (+++)] (D) *C. glandulosum* [Congested blood vessels (++)]; Gliosis (+); Oedema (++)] E: Quercetin [Congested blood vessels (++)]; Gliosis (++)]; No Oedema].

5. Conclusion

On the basis of behavioral studies, and antioxidant effects, marked decrease in cerebral infarct size. Histopathological report, it can be concluded that the hydroalcoholic extract of leaves of *C. glandulosum* exhibit cerebroprotective activity.

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